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Energy Harvesting Based on SLIPT in I2V-VLC System

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***Abstract*—Vehicular connectivity is mostly based on wireless access. The vehicular applications may be limited due to the limited battery life of the equipment involved. To address this issue, a method based on simultaneous light wave information and power transfer (SLIPT) is proposed for harvesting the energy in the Infrastructure-to-Vehicle Visible Light Communication (I2V-VLC) system. The purpose of SLIPT is to harvest energy using light sources while decoding information. This article studies the effect of SLIPT in an I2V-VLC system. In this method, the received light from the traffic light source at the vehicle is harvested while decoding the information signal. First, for I2V-VLC channel modeling, a recent realistic channel model using the ray-tracing method is utilized. Then, we propose the energy harvesting analysis based on the SLIPT strategy. After that, we investigate the impact of both the longitudinal and lateral distance between the vehicle and the traffic light on the amount of harvested energy. Furthermore, we investigate the relationship between the achievable information rate and the harvested energy amount. The obtained results demonstrate the significant impact of the communication distance and the required information rate on the quantity of harvested energy.**

***Index Terms*—Energy harvesting, traffic light-to-vehicle, VLC, SLIPT.**

I. Introduction

Visible light communication (VLC) technology has become a promising candidate for Radio Frequency (RF) technology to achieve the wireless connectivity between the vehicles and road infrastructures. This technology depends based on the capacity to adjust light intensity of the light-emitting-diode (LED) to transmit information while achieving the illumination purpose. Nowadays, LED light sources are widely adapted in modern vehicles and infrastructures. That realizes various vehicular VLC connectivity forms including vehicle-to-vehicle (V2V) based on both headlights and taillights and infrastructure-to-vehicle (I2V) based on both streetlights and traffic lights .

In the near future, electric and connected vehicle will be one of the main elements of smart and green city. However, one of the main challenges that limits the applicability and the adaption of such vehicles is the energy management, which requires the utilization of all resources toward efficient charging management. This prompts the deployment of energy harvesting (EH) techniques in the VLC systems to harness the received optical energy and convert it into an electrical one. Thus, the battery lifetime of electric vehicle can be extended. One of the most promising energy harvesting techniques in VLC systems is the Simultaneous Light Wave Information and Power Transfer (SLIPT) [1], which can provide illumination, transfer information, and energy harvesting, simultaneously. In SLIPT systems, The received light wave's Direct Current (DC) part is utilized for the EH, while the data transmission process uses the Alternative Current (AC) component [2]. Such energy harvesting techniques are extensively studied in the context of indoor VLC applications [3]. For example, in [4], the rate maximization approach for a dual-hop VLC/RF network was examined, where the relay collects energy from the VLC connection using a photodetector (PD) and then re-transmits the data to the RF link.

However, none of these works investigated the performance of SLIPT in the context of vehicular applications. Consequently, we want to close this research gap in our work by investigating the performance of SLIPT technique in an I2V VLC system. In which, the incident light from the traffic light source is harvested while decoding the information signal. We utilize a most recent path loss model for I2V VLC channel modeling based on ray-tracing approach [5] which is a realistic channel model, as VLC in vehicles is based on the need for a more accurate representation of propagation light in the environment. Ray-tracing models realistically simulate the physical behavior of light and take into account factors such as reflections, refractions, and diffraction, which can affect VLC system performance. Then, we propose the energy harvesting analysis based on the SLIPT strategy and investigate the impact of both the longitudinal and lateral shift between

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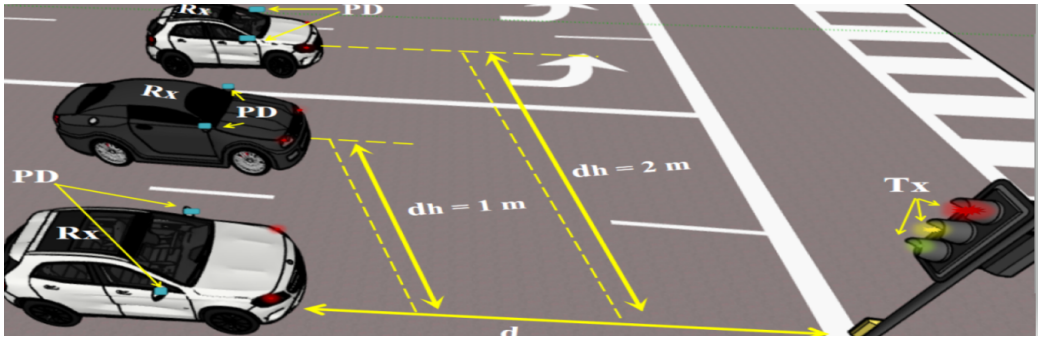

Fig. 1. I2V-VLC system.

the vehicle and the traffic light. We further investigate the relationship between the achievable information rate and the harvested energy amount.

According to the paper's structure, Section II describes the channel model and the system. The performance analysis is illustrated in Section III. Section IV presents and discusses the numerical results. Section V concludes the paper.

II. Model for the System and Channel

A. System Model

We assume a I2V-VLC system on a one-lane road, as shown in Fig. 1 on a roadway where (d) is the separation distance between the traffic light (S) and the destination car (D). We consider a possible lateral shift of dh between the vehicle and the traffic light pole. Each light of the traffic light is used as a wireless VLC transmitters (TX). It has a transmit power of P_l. On the other side, two PDs, located on the top the mirrors of each car, are deployed as wireless VLC receivers (RX). Each has a responsivity of R.

B. Methodologies for Channel Model

Non-sequential ray-tracing is the foundation of our channel modeling methodology. It was previously employed for the creation of an indoor VLC channel [6] and then applied to outdoor one [5]. In this approach, the OpticStudio program is used to build a 3D simulation platform using CAD models of automobiles and roadways. Then, the surface coating material of the CAD object is defined, and the scattering fraction and the wavelength-dependent reflectance are specified. After that, the specifications of the headlight and the taillight, such as optical power, orientations, and radiation pattern are defined. Similarly, the characteristics of the detector such as the field of view angle, aperture diameter, and orientations are specified.

After constructing the 3D platform, the non-sequential ray-tracing model is performed to generate an output file containing the path length and the power of each ray that reached the receiver. These statistics are then imported into Matlab to obtain the channel impulse response (CIR). Suppose k is the total number of rays reaching the photodetector (PD), P_i is the optical power, and τ_i is the propagation time of the ith ray. The DC channel gain is calculated by

$$h = \int_0^{\infty} \{\sum_{i=1}^{k} P_i \delta\left(t - \tau_i\right)\} dt, \tag{1}$$

where δ denotes the Dirac delta function. The path loss is then obtained by [7]

$$PL = 10 \log_{10}\left(H\right) = i \exp\left(-\left(\frac{d-j}{k}\right)^2\right), \tag{2}$$

where for each d_h value under consideration, the values of i, j and k coefficients are given in [7].

III. Analysis Performance

The performance of energy harvesting based on SLIPT in the I2V-VLC system is examined in this section.

A. Optical Transmission Using Wireless Technology

Assuming that the modulated electrical signal representing the information source is denoted by $s(t)$, A DC bias M is applied to $s(t)$ to ensure that the resulting optical signal is not negative. The LED's transmitted optical signal from the traffic light is [2]

$$P(t) = P_l[M + s(t)], \tag{3}$$

where P_l represents the LED transmit power per unit W/A of electrical current $M + s(t)$. The electrical signal varies in relation to the DC bias $M \in [I_{max}, I_{min}]$ with peak amplitude N, where I_{min} and, I_{max} respectively, are the minimum and the maximum input bias currents.

The following limitation is created to minimize clipping distortion caused by the LED's non-linearity by restricting the input electrical signal to the LED inside the linear region of the LED's functioning.

$$N \leq \min\left(M - I_{max}, I_{min} - M\right). \tag{4}$$

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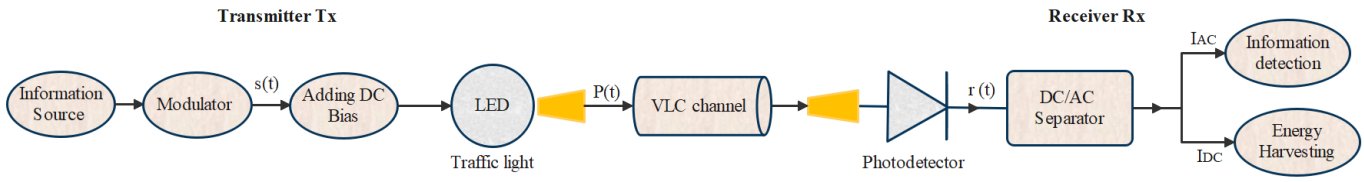

Fig. 2. SLIPT design for I2V-VLC system.

B. Electrical Received

The electrical current $r(t)$ at the output of the PD can be written as

$$r(t) = I_{DC}(t) + I_{AC}(t) + n(t), \tag{5}$$

where $n(t)$ is indeed the additive white Gaussian noise (AWGN), which is made up of thermal noise and background shot noise. $I_DC(t)$ and $I_AC(t)$ are the DC and AC components, respectively.

The AC component $I_{AC}(t)$ is made up of two terms, namely, I_{AC1} (which is due to the dedicated LED) and I_{AC2} (that is due to other interfering sources) as follows

$$I_{AC}(t) = I_{AC1}(t) + I_{AC2}(t), \tag{6}$$

where

$$I_{AC1}(t) = RhP_l s(t). \tag{7}$$

The received signal-to-noise-ratio (SNR) can be written as

$$\text{SNR} = \frac{(RhP_l N_1)^2}{N_p}, \tag{8}$$

where N_p is the noise power, and N_1 is the maximum value of $s(t)$ peak amplitude, with

$$N_1 = \frac{I_{max} - I_{min}}{2}. \tag{9}$$

C. Energy Harvesting

The PD comprises both the $\mathrm{D}C$ and $\mathrm{A}C$ signals. The DC component is stopped by a capacitor and passed via the energy harvesting branch in order to accomplish energy harvesting. Thus, the energy harvested is provided by [8]

$$EH = \eta I_{DC} V_{oL}, \tag{10}$$

where η is the fill factor, V_{oL} being the voltage across the LED, and I_{DC} is the DC component of the output current. It can be given as follows

$$I_{DC} = I_{DC1} + I_{DC2}, \tag{11}$$

$$I_{DC1} = RhP_l M, \tag{12}$$

where I_{DC1} is caused by the dedicated LED and I_{DC2} is caused by other light sources. Also, V_{oL} can be written as

$$V_{oL} = V \ln\left(1 + \frac{I_{DC}}{I_d}\right), \tag{13}$$

where V is the thermal voltage and I_d is the dark saturation current of the PD.

D. SLIPT strategy for I2V-VLC system

In the SLIPT model (as shown in Fig. 2), the intensity of the LED is modulated to transmit information and enable dual use for illumination and communication. Through the VLC channel, an optical signal ($P(t)$) is produced, which is received by the photodetector. After converting the incoming optical signal to an electrical signal, the PD utilizes a time splitter (TS) to separate out the AC and DC components. The DC component is applied to energy harvesting, while the AC component is used for information decoding. As a result, the energy harvesting equation for the TS system is

$$EH_{TS} = Eh_1 + Eh_2. \tag{14}$$

Thus, the communication between Tx and Rx is accomplished in two distinct periods, where

1) period 1: During this phase, the objective is to maximize the received SNR. With a duration of T = [0:1] second. Thus, the quantity of harvested energy is provided for a given value of T.

$$\begin{aligned} Eh_1 =& \eta T \left(RhP_l M_1 + I_{DC2}\right) \\ & V_t \times \ln\left(1 + \frac{RhP_l M_1 + I_{DC2}}{I_d}\right), \end{aligned} \tag{15}$$

where M_1 is the maximum value of the DC bias.

$$M_1 = \frac{I_{max} + I_{min}}{2}, \tag{16}$$

2) Period 2: During this phase 1-T, the aim is to maximize the energy harvested, i.e., N = 0 and M= I_{max}, where

$$\begin{aligned} Eh_2 =& \eta(1-T) \left(RhP_l I_{max} + I_{DC2}\right) \\ & V_t \times \ln\left(1 + \frac{RhP_l I_{max} + I_{DC2}}{I_d}\right). \end{aligned} \tag{17}$$

On the other hand, SLIPT creates a trade-off between the amount of energy harvested and the quality of service (QoS) provided. Therefore, the data rate can be calculated as follows

$$D_R = T \log_2\left(1 + \frac{e}{2\pi}\text{SNR}\right). \tag{18}$$

IV. Numerical Results and Discussion

In this section, we introduce the numerical results of the maximum achievable energy harvested considering system parameters.

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TABLE I
PARAMETERS

Parameters	Values
Responsivity R	0. 4 A/W
Fill factor η	0. 75
Transmit power P_l	20 W/A
Noise power N_p	10^{-15} W/Hz
Longitudinal distance d	35 m
Thermal voltage V	25 mV
Dark saturation current I_d	10^{-6} A
Maximum input bias currents I_{max}	9 to 18 mA
Minimum input bias currents I_{min}	4 to 13 mA
Aperture diameter D_r	1 cm, 2.5 cm, and 4 cm

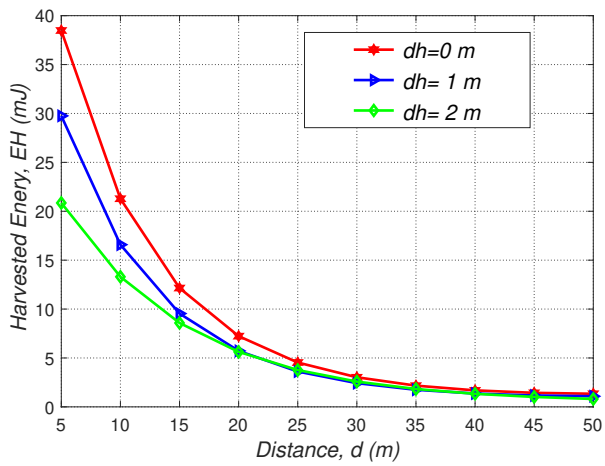

Fig. 3. Maximum energy harvesting versus distance at diverse lateral shift distance

A. Simulation parameters

For the simulation analysis, we assume R = 0.4 A/W, P_l = 20 W/A, N_0 = 10^{-15} W/Hz, d = 45 m, I_{max} = 9 to 18 mA, I_{min} = 4 to 13 mA, V = 25 mA, I_d = 10^{-6} and η = 0.75 W/A. All simulation parameters are given in the Table 1.

We further consider three cases of the lateral distance between the car and the traffic light pole, as follows

- **(a)** We assume that the vehicle moves in the right lane with a perfect alignment with the traffic pole (i.e., dh = 0 m).
- **(b)** We assume that the vehicle moves in the center lane with a misalignment of dh = 1 m from the traffic light.
- **(c)** We assume that the vehicle moves in the left lane, forming a horizontal shift of dh = 2 m with the traffic light.

Fig. 3 shows the maximum energy harvesting while satisfying the target distance, assuming different lateral distance between the cars and the traffic light pole. It is observed that the energy harvested decreases as the distance between the transmitter and the receiver increases. This is because when the transmitter gets farther away from the receiver, the received power that the receiver uses to harvest the energy decreases.

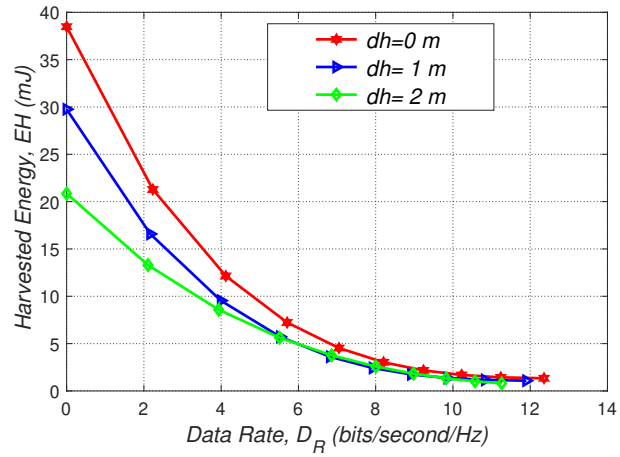

Fig. 4. Maximum energy harvesting versus data rate at diverse lateral shift distance

It is observed also that when the distance between the car and the traffic light is closer, the harvested energy is better. For instance, consider d = 5 m. The maximum energy harvesting for dh = 0 m is given as EH = 38 mJ. This reduces to 30 mJ, and 21 mJ for dh = 1 m, dh = 2 m, respectively.

In Fig. 4, we present the maximum energy harvesting versus data rate assuming different lateral distance between the cars and the traffic light pole. It is observed that the energy harvested decreases with the increase in data rate. This is because the higher the signal, the less energy is recovered. For example, consider D_R = 4 m. The harvested energy for dh = 0 m is EH = 12.5 mJ. This decreases to 9 mJ, and 7 mJ, respectively, for dh = 1 m, dh = 2 m.

V. CONCLUSION

In summary, the paper discussed the potential of the I2V-VLC system based on the SLIPT technique for sustainable energy harvesting in vehicular communication applications. The study revealed that the proposed method, which takes into account the asymmetrical intensity profiles of traffic lights, can significantly increase the battery capacity for vehicles. The results showed that the harvested energy increases as the information rate and the distance between the vehicle and the communication source decrease.

Additionally, the study highlighted the importance of considering the effect of traffic light intensity patterns on the energy harvesting performance, as this can greatly affect the efficiency of the system. The results of this study provide valuable insights for future research in the field of energy harvesting for vehicular communication and can pave the way for the development of more efficient and sustainable energy harvesting systems.

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